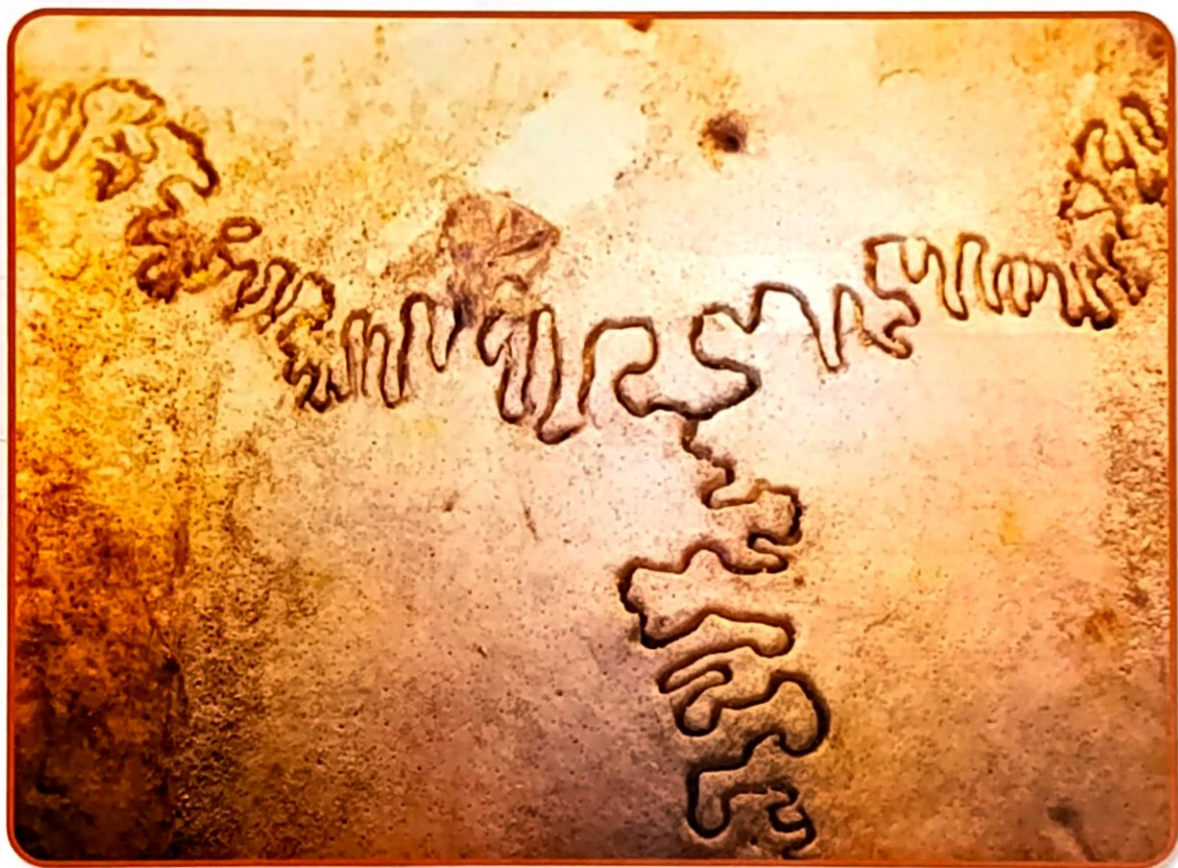


3 Skeleton and muscles

Humans have a skeleton to support their bodies.
What else is a skeleton needed for? Let's find out.



Our skeleton also helps us to move.

Muscles are attached to the bones
of our skeleton.

Joints are places where bones meet.

Joints allow our arms and legs to move in
different directions.

The picture shows the top of a human skull.
The skull protects our brain from damage.

Can you see where the
bones of the skull meet?
Unlike other moveable
joints in our arms and
legs, this joint is fixed.

The human skeleton: support

All vertebrates have **bones** inside their bodies.

These are the bones inside a giraffe.

Each bone in a giraffe's neck is very **long**.

Can you see its **backbone**?

Humans are vertebrates. We call the bones inside our body our **skeleton**.

Our neck bones are much **shorter** than a giraffe's.

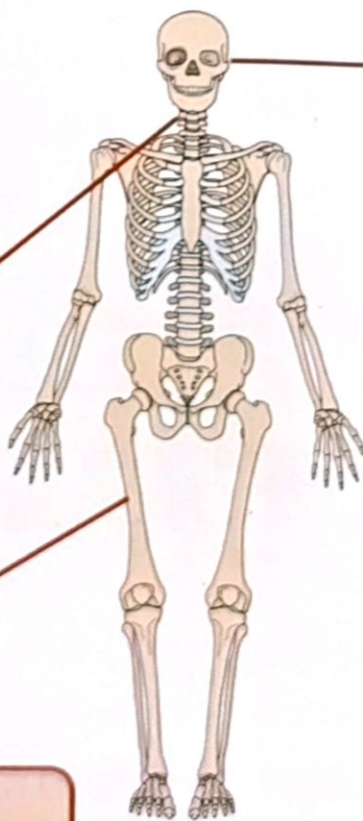
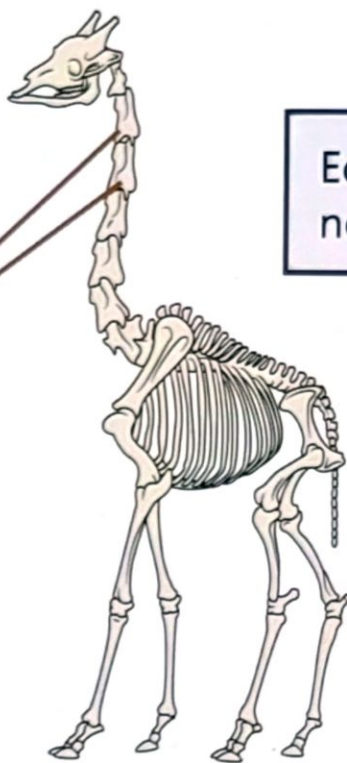
Our smallest bones are inside our ears.

Blood cells are an important part of our blood. They are made inside some of our bones.

This is the longest bone in our skeleton.

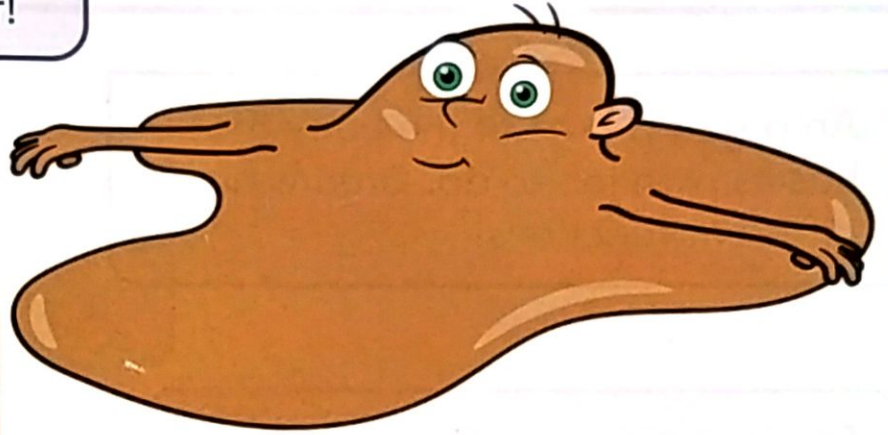
Can you see the small bones of the backbone called **vertebrae**?

The skeleton does not have ears or a nose. They are not made of bone so we can bend them easily.



One function of a skeleton is **support**. It keeps our body shape.

If we did not have a skeleton we would be in a pile on the floor!



Our hands are made of lots of bones.

Compare your own hands with this picture.

Can you work out where some of the bones are? Find the **thumb** first.



Key words

bones

long

backbone

skeleton

shorter

blood cells

vertebrae

support

thumb

The human skeleton: protection

The human body contains many **organs**.

An organ is part of the body that has its own job to do. Organs have different functions.

Put your hand on your chest.
Can you feel your **heart beating**?

Your heart is an organ.
It keeps you alive.

Organs are much softer than bone, so they need to be **protected** from **damage**.



Your **brain** is another organ.
It is inside your head.
It helps you think and do things.

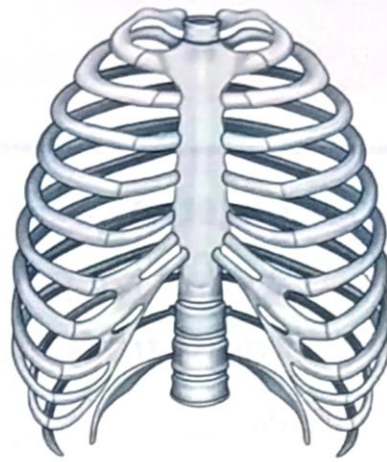


Your **lungs** are also organs.
They help you to **breathe**.

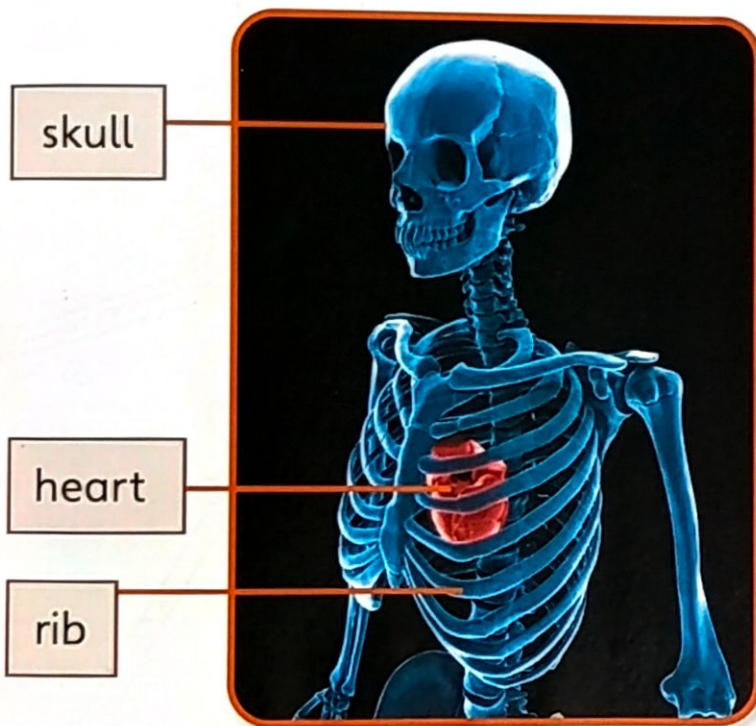


Your **rib cage** protects your heart and lungs.

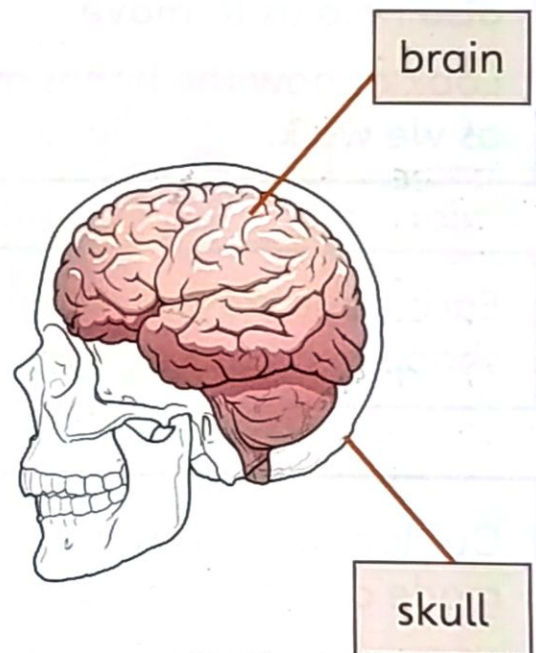
Each of the bones across it is a **rib**.



Look at this skeleton. The heart is drawn inside the rib cage to show you where it is.



Now look at the **skull**.



Key words

organs

heart

beating

protected

damage

lungs

breathe

brain

rib cage

rib

skull

Your soft brain is inside your hard skull. The hard skull protects it from damage.

The human skeleton: movement

The bones of our skeleton:

- support our body and keep it in shape
- protect softer organs from damage
- make blood cells.

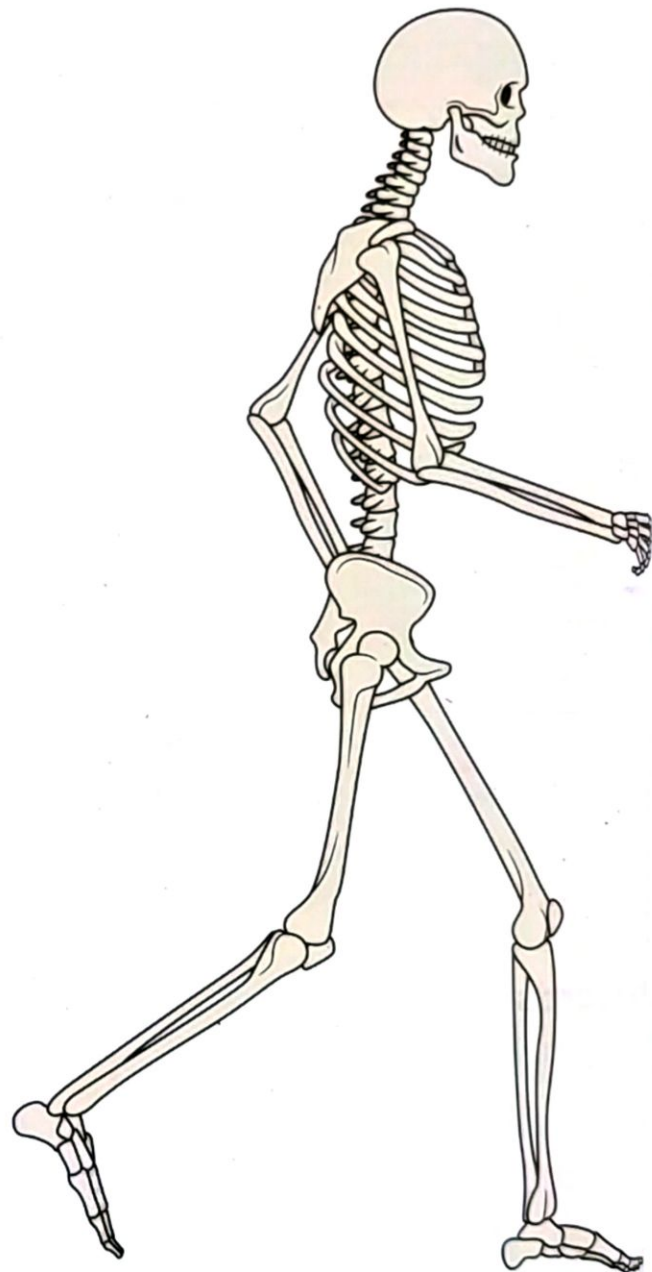
The bones of our skeleton also help us to **move**.

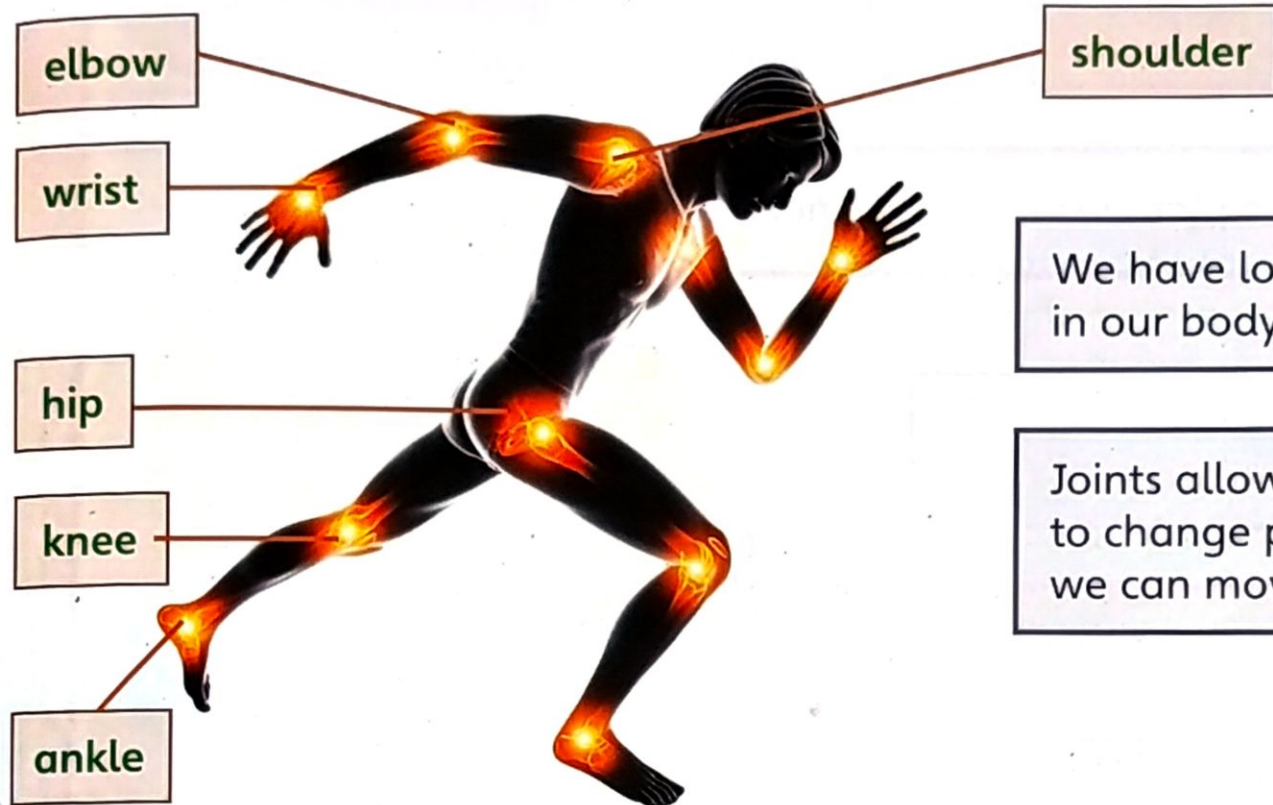
Look at how the bones move as we walk.

Each bone stays the same shape because it is very hard.

Our legs and arms are each made of more than one bone.

The bones **meet** at a **joint**.





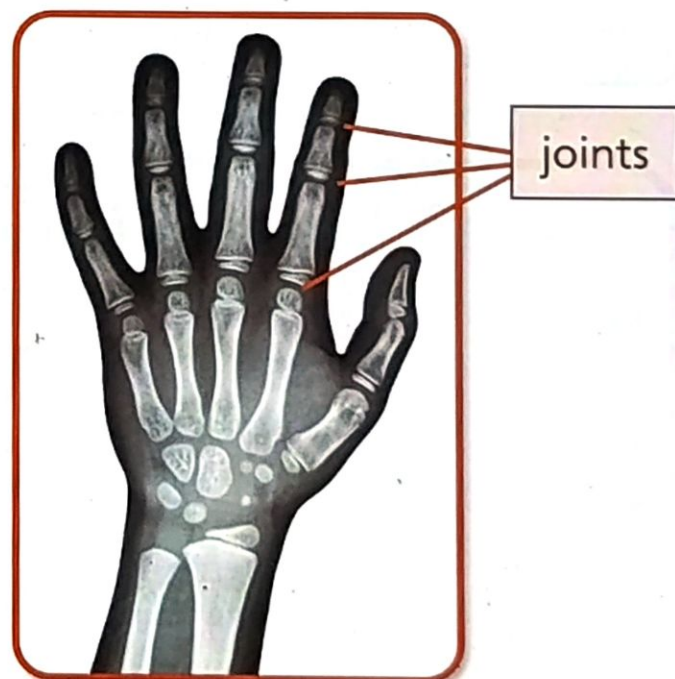
We have lots of joints in our body.

Joints allow our bones to change position so we can move.

This is what the bones of our wrist and hand look like.

Can you see any more joints?

Find these joints on your own fingers. Look at how they help your fingers to move in different ways.



Key words

move

meet

joint

shoulder

elbow

wrist

hip

knee

ankle

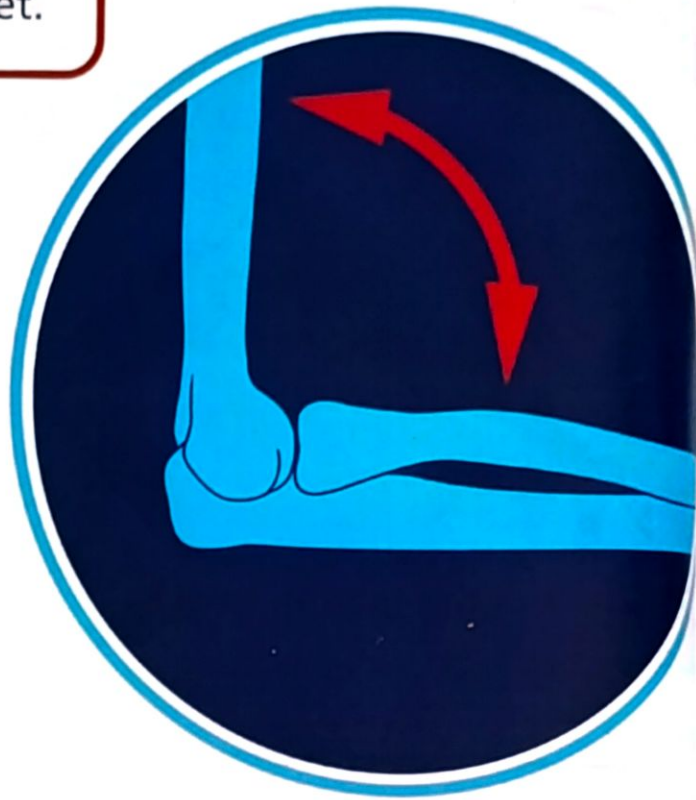
Elbows and knees

A joint is the place where bones meet.

These are the bones in the arm at the elbow joint.

Can you feel these three bones in your arm?

Hold the top of your arm close to your body.



Now move your hand up and down.

How does your elbow move?



Can you touch your shoulder?

Can you make your arm **straight** when you move it down again?

Your knee joint has the same **range of movement** as your elbow joint.

Try this with your lower leg. Keep your hips and upper legs still.

Can you see three long bones at this knee joint too?



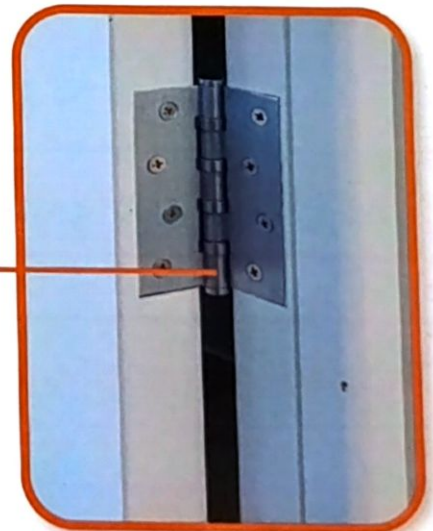
knee cap



The tiny bone is not part of the joint. It is the **knee cap**. It protects the joint.

The elbow and knee move like the **hinge** on a door.

hinge



Key words

straight

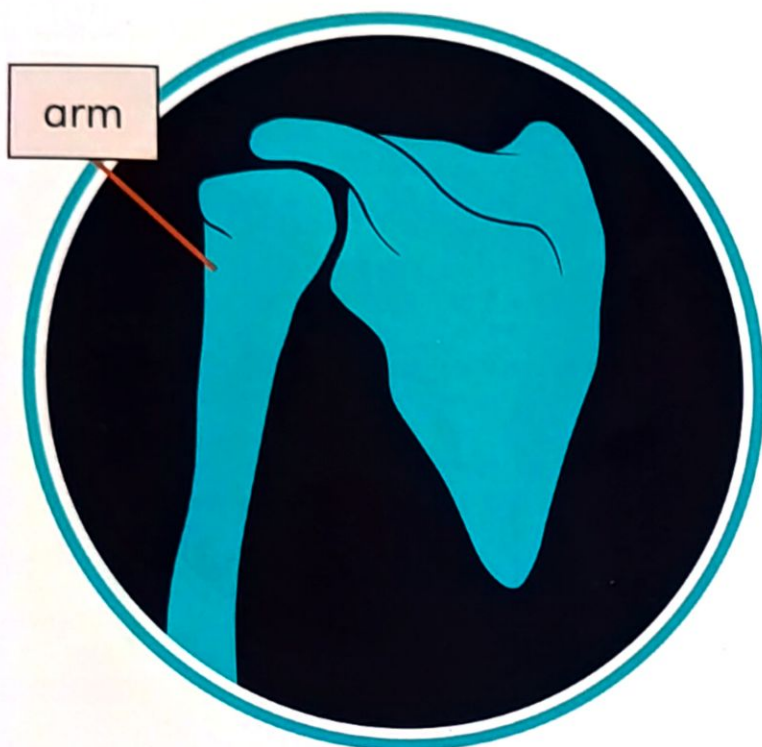
range of movement

knee cap

hinge

More types of joint

Your shoulder joint gives your arm a bigger range of movement than the elbow joint does.



There are two bones at the shoulder joint.

Can you draw a big **circle** in the air using your whole arm?

Find out other ways in which your shoulder joint moves.

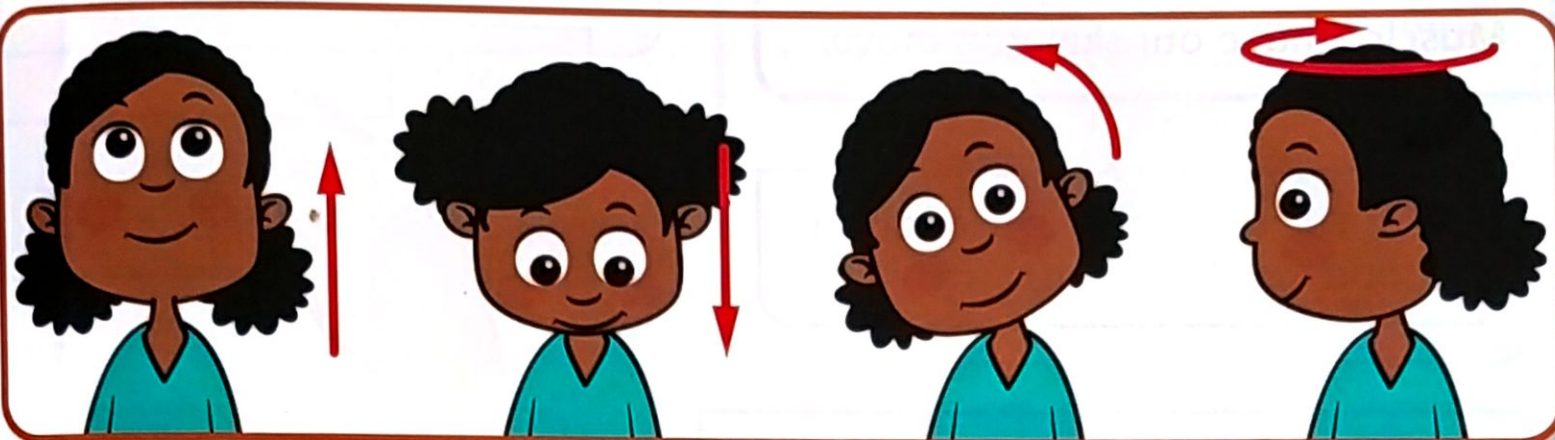
Now try the same with your leg.

The drawing shows two hip joints.

Two bones meet at each hip joint.



Your **neck** joint allows your head to move in many directions.



Some joints do not move. They are **fixed** joints.

This is the top of a human head.

The skull is made of three bones.

When a baby is born, the skull bones are not joined.

As the baby grows, the skull bones join together forever and then do not move.



Key words

circle

neck

fixed

Muscles

Muscles make our skeleton move.

Muscles are red because lots of **blood** goes to them.

Some animals and humans eat meat.
Meat is the muscle of an animal.

Bones cannot move. They need muscles to pull them in different directions.

muscle



bone

This is an elbow joint.

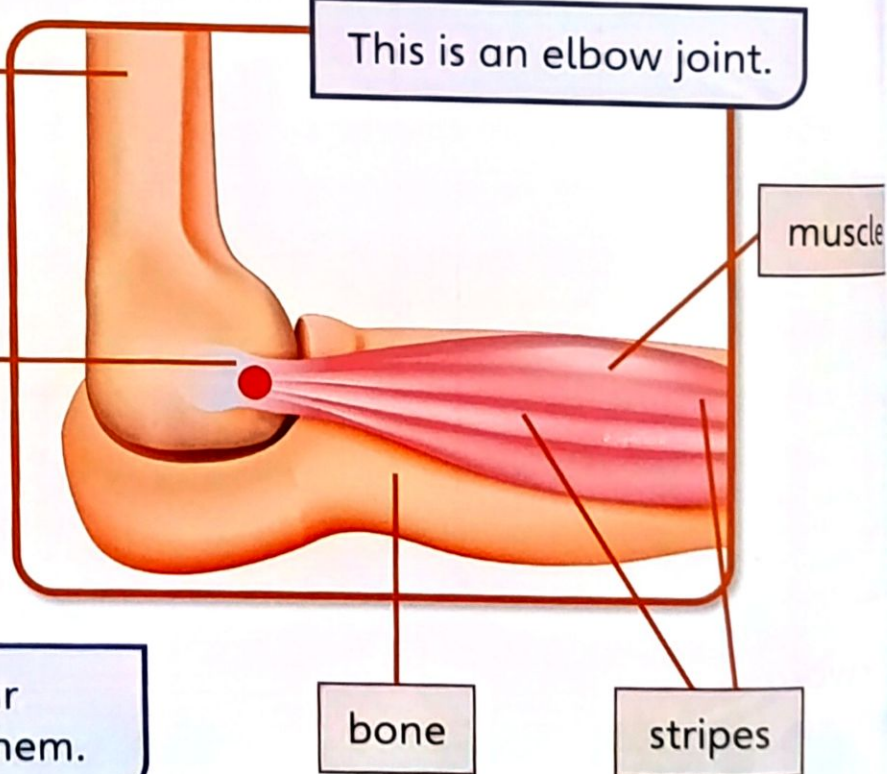
The muscle is **attached** to the bone.

muscle

The muscles that move our skeleton have stripes on them.

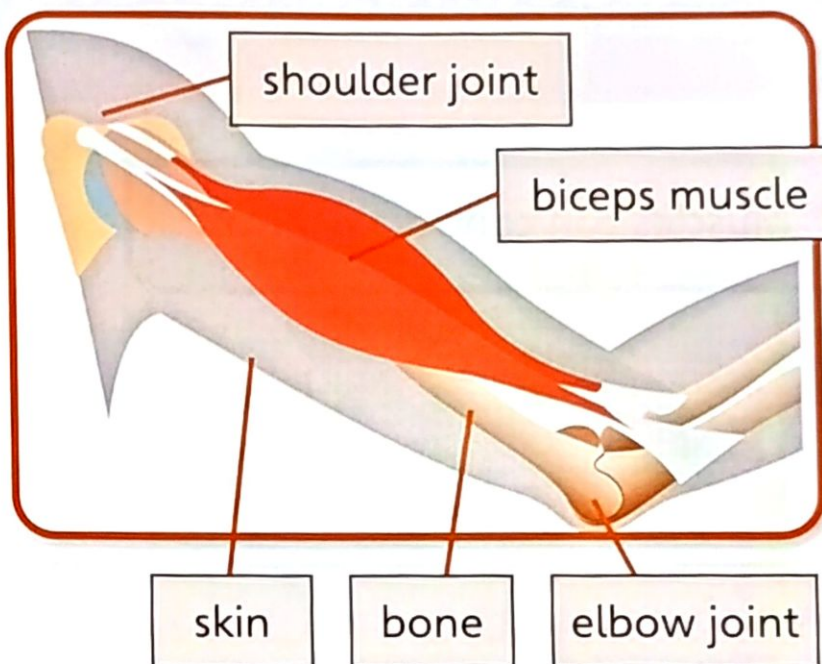
bone

stripes

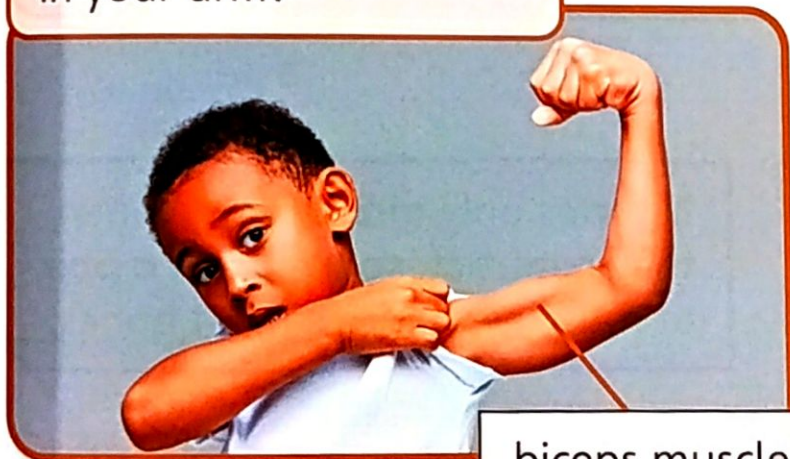


Look at one of the muscles in this arm. It is called the **biceps** muscle.

Each end of this muscle is attached to a different bone.



Can you feel this muscle in your arm?

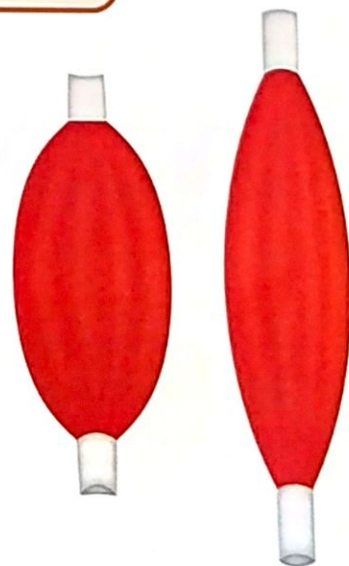


Lift your arm like the child in the picture.

Feel how your biceps changes.

To **bend** the arm, the biceps muscle **contracts**. Now it is shorter and fatter.

To make the arm straight, the biceps goes back to how it started. It **relaxes**.



Key words

muscle

blood

attached

biceps

bend

contracts

relaxes

Contracting and relaxing

Muscles can contract or relax.



When a muscle contracts it becomes shorter and fatter.



When a muscle relaxes it goes back to the length and shape it started.

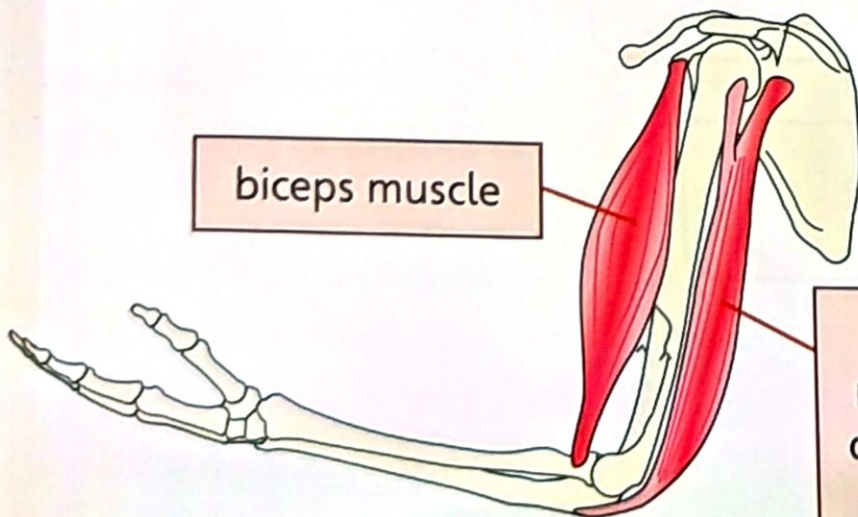


Muscles do not get longer. They stay relaxed or they shorten by contracting.

Muscles only move bones when they contract, so we need a **pair** of muscles to move most bones.

A pair means two that go together, like a pair of socks or a pair of gloves.

biceps muscle



This **triceps** muscle makes a pair with the biceps.



Try doing this movement with your arm.

Can you feel the muscles change?

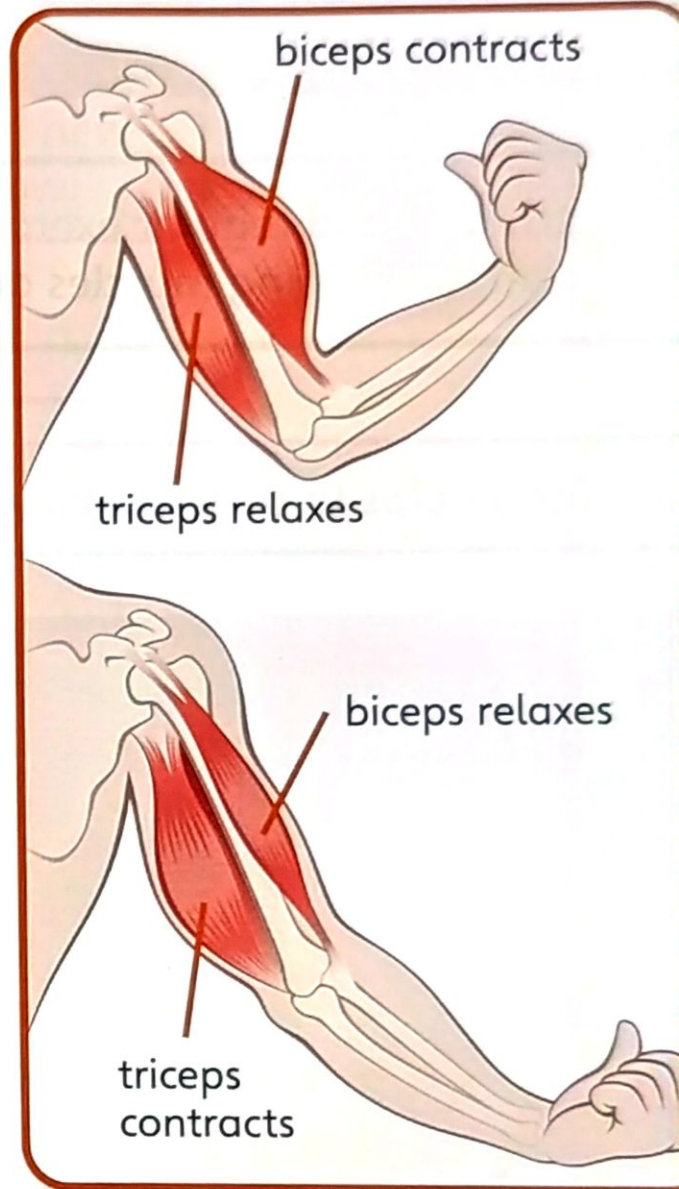
The muscle that is contracting pulls the bone it is attached to.

There are pairs of muscles in your legs too.



Bend and **straighten** your foot.

Can you feel the muscles in your lower leg changing?



Key words

shorten

pair

triceps

straighten

Healthy muscles and bones

A healthy diet and regular exercise are both important for keeping muscles and bones healthy.

Dairy food helps to build growing bones.



So do these other foods.

What can you see here?

Exercising outdoors in sunlight helps to keep bones strong.



The sunlight helps us to absorb the things our bones need from our food.



Protein foods in our diet help to build new muscle as we grow.



Different sorts of exercise use different muscles.



We use lots of muscles to throw a ball.

Karate uses different muscles.



Stretching our body is good for our muscles too.



This girl is improving her **balance**.



Key words

dairy

protein

stretching

balance